

Aishwarya Krishnakumar

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Education

University of Illinois Urbana-Champaign
Master of Computer Science

Fall 2026 - Expected Dec 2027

Sri Sivasubramaniya Nadar College of Engineering
B.E. - Computer Science & Engineering.

Aug 2020 – May 2024

CGPA: 9.58/10, **Outstanding Scholarship winner** for 1st & 2nd year academics. (Ranked 2nd in a cohort of 140)

IIT Madras

Sept 2020 – Apr 2023

B.S - Programming and Data Science.

Completed Foundation Level, earned 78 credits across Data Science and Programming Diploma coursework.

Experience

Citi

Jul 2024 – Present

Tech Program App Dev

Chennai

- Developing and maintaining **middleware and core backend services** for payments, customer accounts, and CRS (Citi Retail Services) using **Java** and **Spring Boot** within a microservices architecture in an Agile environment.
- Designed and Developed **5 new microservices** across payments and promotions domains while managing deployment, proactive monitoring, health checks, security remediation and **production issue resolution**.
- Leading a team of 4 engineers** in the promotions domain, successfully delivering **3 large-scale feature** rollouts across customer-facing microservices.
- Driving **integration efforts for new APIs** and enhancements to existing functionalities, supporting seamless onboarding and operational continuity across customer services, **improving service efficiency by 35%**.
- Won **multiple internal hackathons** and **presented at AI tech conferences** for building GenAI- and LLM-driven solutions, including LexiTest-AI and an MCP-based Agentic AI orchestration platform, collectively reducing development and testing effort by up to **70%**.
- Role:** Backend Developer with hands-on experience in Java, Spring Boot, OpenShift, Jenkins, and RLM, specializing in microservice design, development, deployment, and optimization.
- Consistently recognized with **4+ quarterly performance excellence awards** for high-impact contributions.

FaceTagr

Jan 2024 – Jun 2024

Data Analyst Intern

Chennai

- Optimized **large-scale data** pipelines, improving **process efficiency by 30%** and reducing latency.
- Performed Exploratory data analysis (**EDA**) to uncover business-critical insights, enabling faster and **smarter data driven decisions**.
- Built and deployed machine learning models **predicting customer behavior**, boosting engagement **by 15%**.
- Applied **image processing** and **OCR technologies** to extract insights from visual data and expand analytics reach.

Citi

May 2023 – Jul 2023

EIO&T Summer Analyst Intern

Chennai

- Implemented a robust **Resource Management** application and enabled utilization tracking, leading to increased project efficiency within the **Agile SDLC** framework.
- Created a seamless **bulk upload** feature for resources and vendor data, **reducing manual entry time by 60%**.
- Introduced a **Dynamic Scrum Member Threshold mechanism**, preventing resource overallocation.
- Role:** Designed, developed, and tested the front-end, back-end, and APIs. Played an additional role of Scrum Master.

Projects

Agentic AI for Intent driven API Orchestration (Citi) MCP| Agentic AI| LLM| Python

Apr 2026

Built an MCP-based Agentic AI platform for intent-driven orchestration of payment workflows using natural language inputs. Integrated tool selection and multi-step API execution to automate 25+ complex business processes.

LexiTest AI - LLM Powered API Testing & Workflow Validation (Citi) GenAI | LLMs| Java | Spring Boot

June 2025

Implemented AI-driven validation of customer account scenarios and API workflows using Generative AI and LLMs. Automated workflow validation and execution reporting, reducing development and testing effort by up to 70%.

AI-Powered GST Reconciliation and eBRC Automation (Citi) GenAI | LLMs| Python | Uvicorn

Jan 2025

Automated tax reconciliation workflows, reducing processing time by 45% and eliminating manual government portal interactions. Built AI-driven correlation and reconciliation engines, alongside a conversational assistant for querying reports and extracting insights.

Real-time Tamil Dialect Speech Recognition and Summarization  *Deep Learning/ Data Processing* Jan 2024
Fine-tuned speech recognition models on 45,000+ recordings spanning 6 Tamil dialect variations, extending recognition beyond standardized speech. Enabled real-time transcription and summarization of Tamil speech.

Investment Portfolio Tracker and Analytics Platform *Java/ Springboot/ React JS/ MySQL* Aug 2023
Engineered a full-stack investment management platform with portfolio tracking, transaction management, and performance analytics. Built interactive dashboards for investment monitoring and portfolio evaluation.

Summarizing Indian languages using Machine Learning  *Python/Scikit-Learn/Pytorch/Tensor flow* Oct 2022
Developed a multilingual summarization solution for the FIRE'22 ILSUM competition, processing and summarizing 10,000+ news articles across Hindi, English, and Gujarati languages.

E-commerce Shopper Behavior Analysis: Leveraging ML Models (IIT Madras)  *Python/ Data Processing* Mar 2023
Leveraged predictive models to analyze shopper behavior and purchasing patterns using customer interaction data. Applied machine learning techniques to identify key drivers influencing customer preferences and engagement.

Business Data Management and Analysis (IIT Madras)  *Data Analytics/ Data Visualization/Excel* Mar 2022
Performed EDA on 22,000+ sales records from India's 2nd largest soft-drink distributor. Identified revenue trends, demand patterns, and inventory optimization opportunities through dashboards and data visualizations.

Skills

Languages: Java, Python, C/C++, JavaScript, SQL, Bash

Frameworks Technologies: Spring Boot, Microservices, REST APIs, ReactJS, Node.js, AngularJS

AI/ML: Generative AI, LLMs, Agentic AI, MCP, Scikit-learn, PyTorch, Data Processing, OCR

Cloud, DevOps Tools: AWS S3, OpenShift, Kubernetes, Docker, Unix/Linux, Jenkins, RLM, Lightspeed, Harness

Databases Analytics: MySQL, PostgreSQL, MongoDB, Tableau, Jira, Bitbucket, Git

Publications

Real-time Tamil Dialect Speech Recognition and Summarization  *Circuits, Systems & Signal Processing* Apr 2024
Built a diverse, large-scale Tamil dialect dataset, annotating regional, colloquial, and cultural variations to train real-time ASR models like OpenAI's Whisper and NVIDIA's NeMo. Ensured high-quality data through rigorous validation, enabling improved ASR accuracy and real-time speech recognition performance & published in Springer.

Summarizing Indian languages using Machine Learning  *ILSUM FIRE 2022, CEUR-WS* May 2023
Text summarization for Indian languages (Hindi, Gujarati) and English using pre-trained models as a part of Forum for Information Retrieval Evaluation 2022. Applied mT5-m2m-CrossSum, BERT, XL-Sum, mT5-small and accomplished performance analysis.

Co-Curricular activities

Google Developer Student Club (GDSC), SSN Coding Club (AI/ML Team) - Core Committee Member, IEEE ComSoc - Key Member, Organized multiple events at Invente (SSN's Tech Fest), Youth Red Cross (YRC) - community service volunteer , Quarterly in-house school magazine (Voices) @ SBOA - Sub-Editor.

Certifications

Building AI; Hands-on ML course - **University of Helsinki**, MongoDB Vector Search and MongoDB Aggregation Fundamentals - **MongoDB**, Introduction to Virtual Reality – **Lancaster University**, Designing for a sustainable future – **Samsung**, Presenting Data, Effective Presentation & Effective Leadership – **Hp Life Online Courses**, Introduction to Edge AI - **Intel**, Data Mining - **NPTEL**, Introduction To Industry 4.0 and Industrial Internet of Things - **NPTEL**, Discrete Structures - **NPTEL**